

Lecture 18 Curl Test

Recap. $\vec{F} = \nabla f$

$$\nabla \times \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \times (F_x \vec{i} + F_y \vec{j} + F_z \vec{k})$$

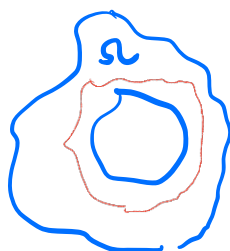
$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}.$$

Def. (Simply-Connected Regions) A region Ω is simply-connected if Ω is connected and every closed loop in Ω can be contracted to a point continuously without leaving the region Ω .

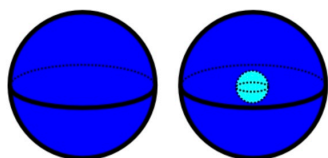


Simply connected

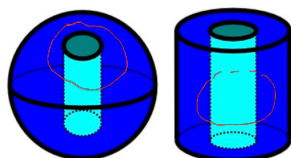


connected, but not simply connected

Simply connected



Non-simply connected



Remarks. $\mathbb{R}^2 / \{0,0\}$ is not a simply connected region

$\mathbb{R}^3 / \{(0,0,0)\}$ is a simply connected region.

Theorem (Curl Test) Given a vector field $\vec{F}$ defined on a region Ω , then 1. If $\vec{F} = \nabla f$ for some scalar function f defined on Ω , then $\nabla \times \vec{F} = \vec{0}$ on Ω

2. If $\nabla \times \vec{F} = \vec{0}$ and Ω is simply-connected, then $\vec{F} = \nabla f$ for some scalar function f defined on Ω .

Pf. Part 1. $\vec{F} = \nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \vec{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \vec{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \vec{k} \\ &= 0 \vec{i} + 0 \vec{j} + 0 \vec{k} \\ &= \vec{0} \end{aligned}$$

Fact. $\nabla \times (\nabla f) = \vec{0}$ for any scalar-valued functions $f \in C^1$

Example, $\vec{F}(x, y) = -\frac{y}{x^2+y^2} \vec{i} + \frac{x}{x^2+y^2} \vec{j}$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$

$$\begin{aligned} \nabla \times \vec{F} &= \left(\frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{x^2+y^2} \right) \right) \vec{k} \\ &= \left(\frac{x^2+y^2-2x^2}{(x^2+y^2)^2} + \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} \right) \vec{k} \\ &= \vec{0} \end{aligned}$$

Unit circle C : $\vec{r}(t) = (\cos t) \vec{i} + (\sin t) \vec{j}$ $0 \leq t \leq 2\pi$

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \oint_C \frac{-y dx + x dy}{x^2+y^2} \\ &= \int_0^{2\pi} \frac{-\sin t (d \cos t) + \cos t (d \sin t)}{\sin^2 t + \cos^2 t} \end{aligned}$$

$$= \int_0^{2\pi} \frac{\sin^2 t + \cos^2 t}{\sin^2 t + \cos^2 t} dt$$

$$= 2\pi \neq 0 \quad \Rightarrow \quad \vec{F} \text{ is not conservative.}$$

Example. $\vec{F}(x, y, z) = -\frac{x\vec{i} + y\vec{j} + z\vec{k}}{(x^2 + y^2 + z^2)^{3/2}}$ on $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$.

$$\nabla \times \vec{F} = \vec{0} \quad \& \quad \mathbb{R}^3 \setminus \{(0, 0, 0)\} \text{ is simply connected} \Rightarrow \vec{F} \text{ is conservative.}$$

Example. Let C be the path in $\mathbb{R}^2$ parametrized by:

$$\vec{r}(t) = (\cos^4 t, t) \vec{i} + \frac{2t}{\pi} \vec{j} \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

Find the line integral:

$$\int_C (2xe^{xy} + x^2 ye^{xy}) dx + (x^3 e^{xy} + 2y) dy$$

Sol. $\vec{F}(x, y) = (2xe^{xy} + x^2 ye^{xy}) \vec{i} + (x^3 e^{xy} + 2y) \vec{j}$ on $\mathbb{R}^2$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xe^{xy} + x^2 ye^{xy} & x^3 e^{xy} + 2y & 0 \end{vmatrix}$$

$$= \left(\frac{\partial (x^3 e^{xy} + 2y)}{\partial x} - \frac{\partial (2xe^{xy} + x^2 ye^{xy})}{\partial y} \right) \vec{k}$$

$$= (3x^2 e^{xy} + x^3 ye^{xy} - 2x^2 e^{xy} - x^2 e^{xy} - x^2 ye^{xy}) \vec{k}$$

$$= 0 \vec{k}$$

$$= \vec{0}$$

$$\Rightarrow \vec{F} \text{ is conservative.}$$

$$\vec{r}\left(-\frac{\pi}{2}\right) = (\cos(-\pi/2))^{24601} \vec{i} - \vec{j} = -\vec{j}$$

$$\vec{r}\left(\frac{\pi}{2}\right) = (\cos(\pi/2))^{24601} \vec{i} + \vec{j} = \vec{j}$$

Path γ from $(0, -1)$ to $(0, 1)$:

$$\begin{aligned}\vec{r}_\gamma(t) &= \langle 0, -1 \rangle + t(\langle 0, 1 \rangle - \langle 0, -1 \rangle) \\ &= \langle 0, 2t - 1 \rangle \quad 0 \leq t \leq 1.\end{aligned}$$

$$\vec{r}'_\gamma(t) = \langle 0, 2 \rangle$$

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_\gamma \vec{F} \cdot d\vec{r} = \int_0^1 \langle 2xe^{xy} + x^2ye^{xy}, x^3e^{xy} + 2y \rangle \cdot \langle 0, 2 \rangle dt \\ &= \int_0^1 2(x^3e^{xy} + 2y) dt \\ &= \int_0^1 4(2t-1) dt \\ &= 4[t^2 - t]_0^1 \\ &= 0.\end{aligned}$$

If $\vec{F}$ is conservative, then 1) $\nabla \times \vec{F} = \vec{0}$

2) $\oint_C \vec{F} \cdot d\vec{r} = 0$ for any closed C

Green's Theorem.

Key question: Is there any hidden relation between $\nabla \times \vec{F}$ and $\oint_C \vec{F} \cdot d\vec{r}$?

Def (Simple Closed Curves) A curve C is called a simple closed curve if the two endpoints coincide and it does not intersect itself at any other point.



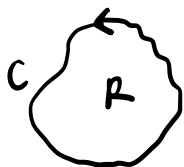
simple closed curve



closed curve, but not simple.

Theorem (Green's Theorem) Let C be a simple closed curve in $\mathbb{R}^2$, which is counter-clockwise oriented. Suppose the curve C encloses region R . Let $\vec{F}(x,y)$ be a vector field which is defined and $\nabla \times \vec{F} \neq \vec{0}$ at every point in R , then

$$\underbrace{\oint_C \vec{F} \cdot d\vec{r}}_{\text{line integral}} = \underbrace{\iint_R (\nabla \times \vec{F}) \cdot \vec{k} dA}_{\text{double integral}}.$$



$$\vec{F} = F_x \vec{i} + F_y \vec{j}, \quad \nabla \times \vec{F} = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}.$$

$$\oint_C F_x dx + F_y dy = \iint_R \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA.$$

Remark: If $\nabla \times \vec{F} = \vec{0}$ and $\vec{F}$ is defined on $\mathbb{R}^2$,

then $\oint_C \vec{F} \cdot d\vec{r} = \iint_R \vec{0} \cdot \vec{k} dA = 0$ for any simple closed curve C